

LEGO: Synergizing Expert GraphRAG and Expert Chain-of-Thought for Legal Reasoning

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Motivation: textual similarity is not legal relevance

Case

Company A required goods to be **transported** to Company B and arranged the **transportation** by telephone with a **transport** company. During **transportation**, the driver negligently caused an accident, damaging the goods and delaying **transportation**, resulting in Company B's loss.

Question From whom should Company B claim compensation for its loss?

A. Company A	**✓ correct**
B. Person C	
C. Driver	
D. The transport company	**✗ similarity**

"Transport" appears 5 times and pulls models toward D, but B has no contract with the carrier: by privity (Art. 465), B must claim from A.

Motivation: two bottlenecks in legal AI

1 Retrieval

- RAG and GraphRAG rank provisions by lexical or semantic similarity
- They overlook the normative relations among provisions: prerequisites, exceptions, general–special

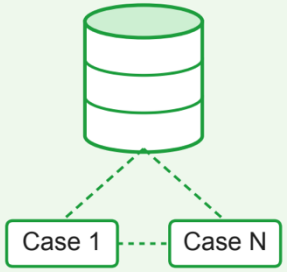
2 Reasoning

- Vanilla CoT can produce plausible rationales but not grounded in retrieved evidence
- They ignores reasoning logic structure, so it is unstable and hallucinates

LEGO injects expert legal structure into both retrieval (ExpertGraphRAG) and reasoning (ExpertCoT).

Method: input case and conceptual fact graph

Multihop Case



Multihop within facts and provisions

Company A required goods to be transported to Company B and arranged the transportation by telephone with a transport company. During transportation, the driver negligently caused an accident, damaging the goods and delaying transportation, resulting in Company B's loss.

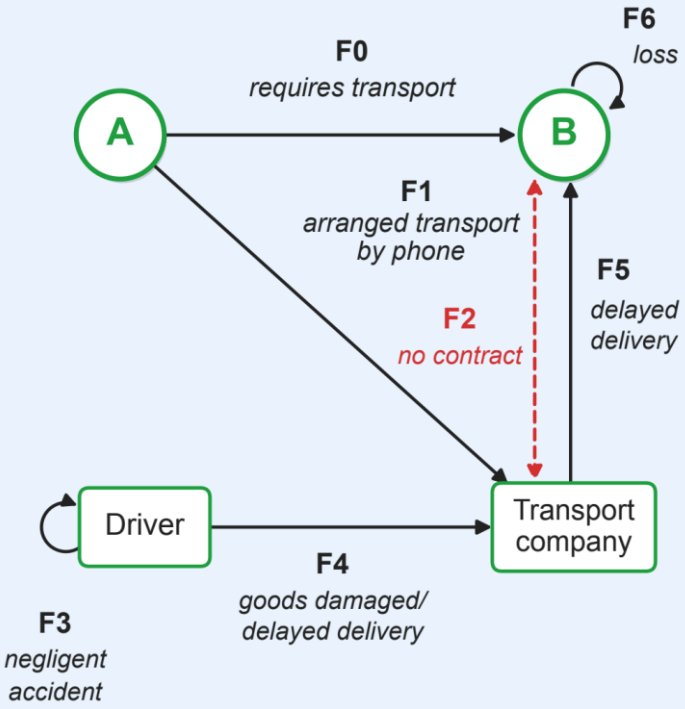
Q **Question:** From whom should Company B claim compensation for its loss?

A. Company A ✓	C. Driver
B. Person C	D. The transport company ✗

Query
case facts + question + options

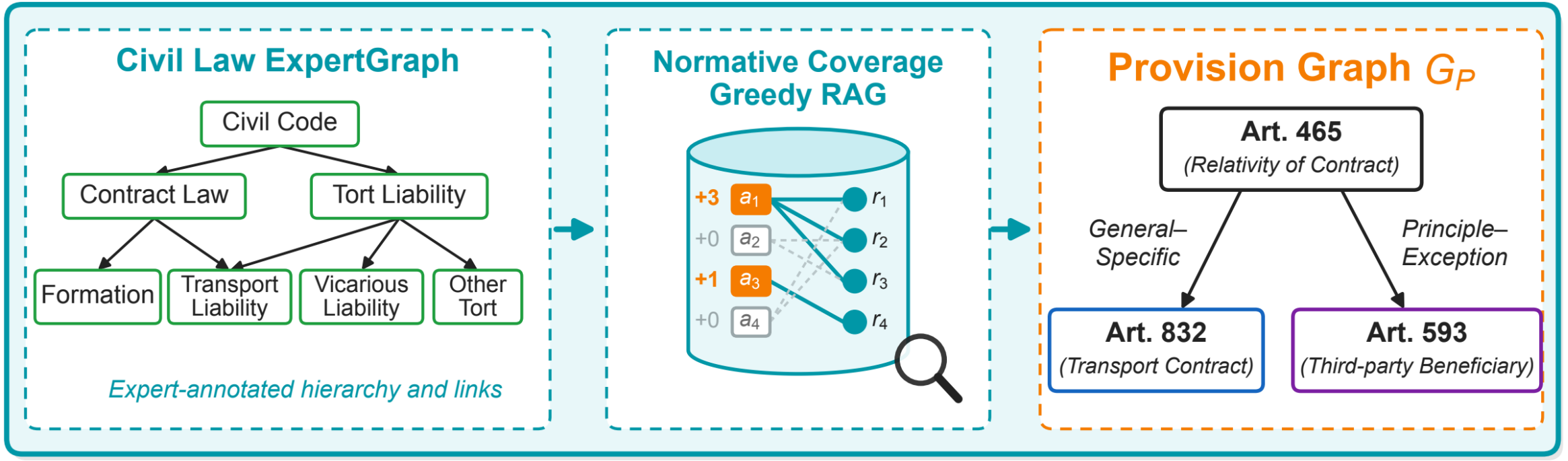
Fact graph G_F
parties and facts F0–F6

Fact Graph G_F



Conceptual only: the fact graph is not materialised; its facts become the minor premises of ExpertCoT

ExpertGraphRAG: from expert graph to case-specific provision graph



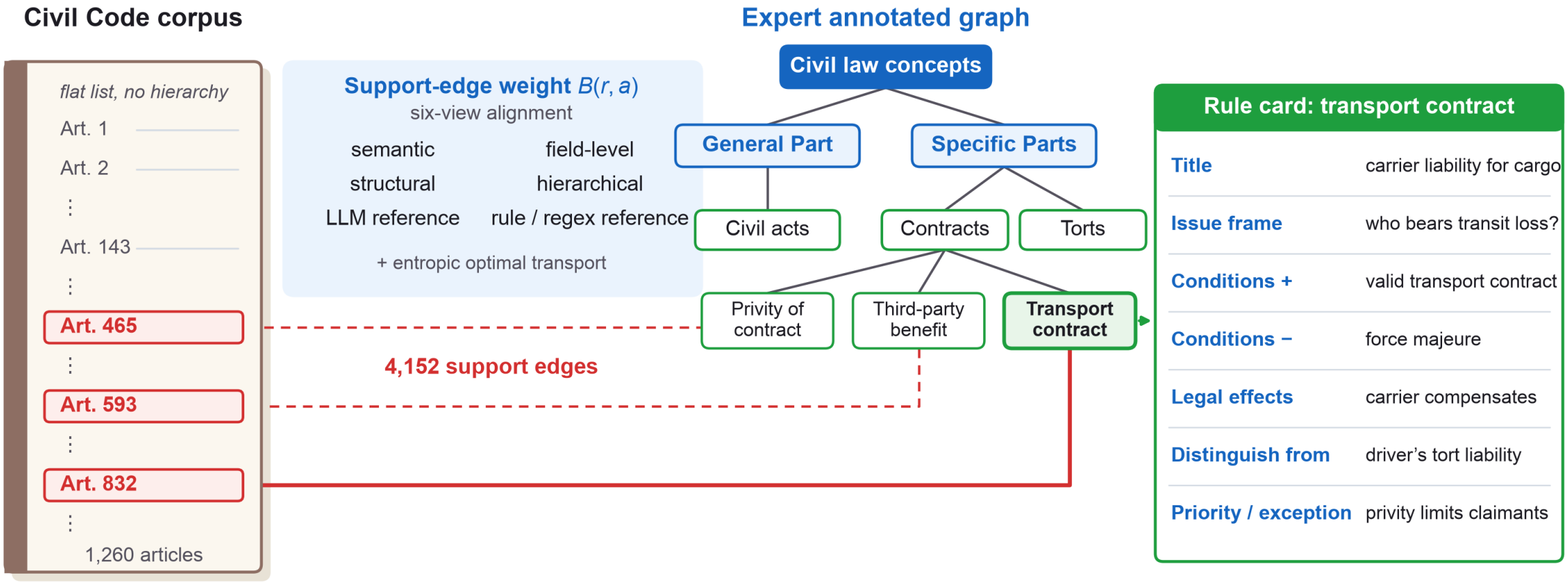
Input
query + expert graph over all 1,260 articles

NCG retrieval
greedy selection by marginal legal coverage (K = 8)

Output
case-specific provision graph with typed relations

ExpertGraphRAG: the Civil Law ExpertGraph

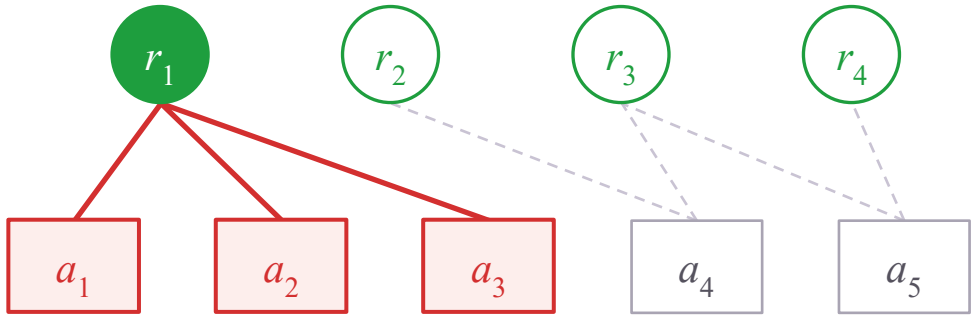
318 legal concepts · **409** rule cards · **4,152** support edges · **1,211** of 1,260 Civil Code articles linked



ExpertGraphRAG: Normative Coverage Greedy (NCG)

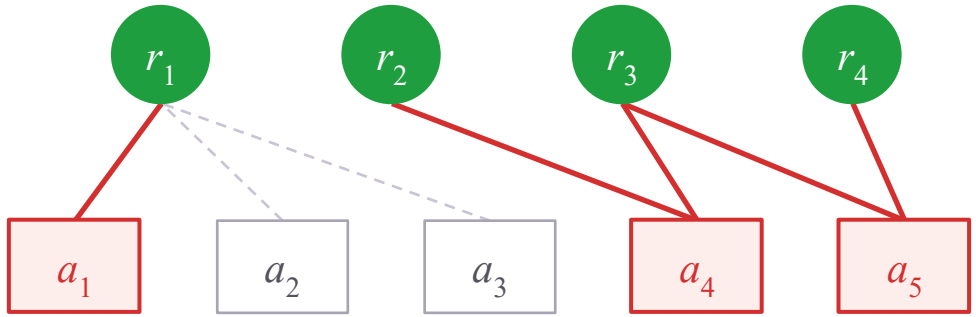
○ legal issues (rule units) □ candidate articles budget: 3 articles

Similarity top-3



1 of 4 issues covered

NCG



4 of 4 issues covered

$$\Delta(a \mid S) = d(a) + \lambda_{\text{align}}\, g(a) + \lambda_{\text{cov}}\, |\Gamma(a) \setminus \Gamma(S)| - \lambda_{\text{red}}\, |\Gamma(a) \cap \Gamma(S)|$$

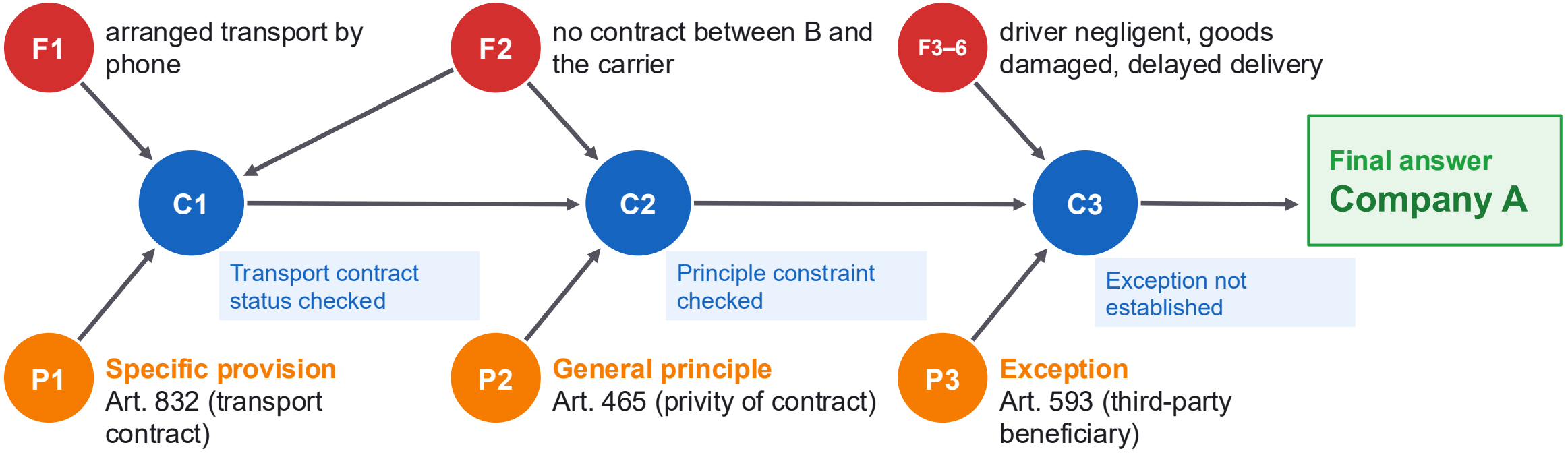
relevance · expert alignment · **newly covered legal rules** · redundancy; greedy: add argmax Δ until |S| = K

ExpertCoT: reasoning as a legal syllogism

P · Provision
major premise: controlling rules, only from the retrieved provisions

F · Fact
minor premise: legally relevant facts, no legal conclusions yet

C · Conclusion
judgment per option: subject, predicate, elements, legal effect



11 explicit instructions, refined through expert-in-the-loop error analysis, all in a single generation pass

Experimental setup

Datasets (civil law)

- **LawExamQA_Civil**: 723 NJEC questions; hops = provisions cited in the official rationale (342 / 215 / 104 / 62)
- **LexRAG_Civil**: open-ended, multi-turn legal consultation
- **PLawBench_Civil**: rubric-based practical case analysis

Metrics

- Exact-match accuracy and set-level F1 (LawExamQA)
- LexRAG: five rubric dimensions + overall, Qwen3-30B-A3B judge
- PLawBench: reasoning + overall, Gemini-3.0-Pro judge; Recall@k for retrieval

Baselines

- **General LMs**: GPT-5, DeepSeek-V3, Qwen3-30B-A3B, GLM-4.7-Flash, Qwen3-8B, GLM-4-9B
- **Legal LMs**: DISC-LawLLM, LegalOne
- **RAG**: dense top-k, LightRAG, RAPTOR, G-Retriever, HippoRAG 2
- **CoT**: Zero-shot-CoT, IRAC-CoT

Implementation

- Same Qwen3-8B reader and Civil Code corpus for all RAG systems
- Greedy decoding (T = 0), thinking mode off; NCG with K = 8

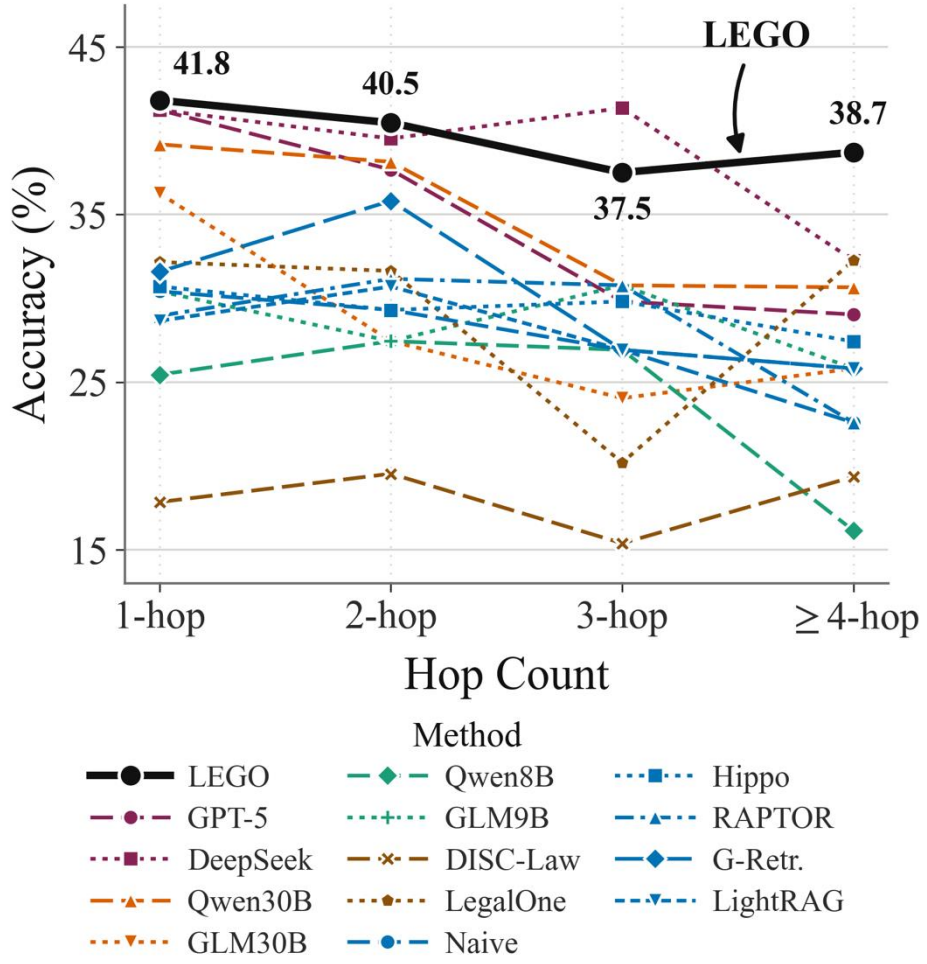
Results: LEGO (8B) beats RAG baselines and much larger LLMs

Method	Size	All	1-hop	2-hop	3-hop	≥4
Closed-source & large general LMs						
GPT-5	API	37.48	41.23	37.67	29.81	29.03
DeepSeek-V3	671B	39.97	41.23	39.53	**41.35**	32.26
Open larger LMs						
Qwen3-30B-A3B	30B	36.93	39.18	38.14	30.77	30.65
GLM-4.7-Flash	30B	30.98	36.26	27.44	24.04	25.81
Small open-source LMs						
Qwen3-8B	8B	25.45	25.44	27.44	26.92	16.13
GLM-4-9B-chat	9B	29.18	30.41	27.44	30.77	25.81
Legal-domain LMs						
DISC-LawLLM	7B	18.12	17.84	19.53	15.38	19.35
LegalOne	8B	30.29	32.16	31.63	20.19	32.26
RAG baselines (same Qwen3-8B reader and corpus)						
Naive RAG	8B	28.91	30.41	29.30	26.92	22.58
HippoRAG 2	8B	29.88	30.70	29.30	29.81	27.42
RAPTOR	8B	29.32	28.95	31.16	30.77	22.58
G-Retriever	8B	31.67	31.58	35.81	26.92	25.81
LightRAG	8B	28.77	28.65	30.70	26.92	25.81
LEGO (ours)	**8B**	**40.53**	**41.81**	**40.47**	**37.50**	**38.71**

- **40.53%** overall with an 8B reader, best of all systems
- +8.9 pp over the best RAG baseline (G-Retriever) with the same reader and corpus
- Above GPT-5 and DeepSeek-V3 (671B), and above legal-domain LMs

LawExamQA_Civil, exact-match accuracy (%), 723 items (342 / 215 / 104 / 62 by hop count). Bold = best, underline = second best.

Results: LEGO stays stable as the number of hops grows

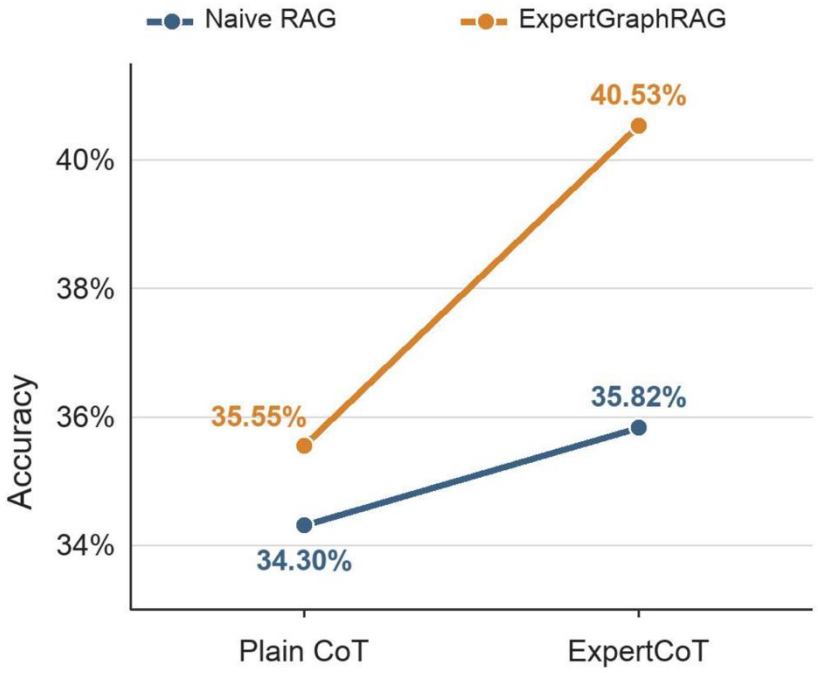


- **LEGO**: from 41.8 (1-hop) to 38.7 (≥4-hop), best on ≥4-hop questions
- **GPT-5**: from 41.2 to 29.0, a drop of 12.2 pp
- Most baselines degrade sharply on 3-hop and ≥4-hop questions

Results: the two modules are complementary

Model	Configuration	Acc	F1	ΔAcc	ΔF1
No-RAG baselines					
Qwen3-8B	zero-shot	0.2545	0.5382	↑ 0.1508	↑ 0.0305
Qwen3-8B	CoT	0.2918	0.4728	↑ 0.1135	↑ 0.0959
Qwen3-8B	IRAC-CoT	0.3112	0.4973	↑ 0.0941	↑ 0.0714
RAG + standard CoT					
Qwen3-8B	Naive RAG + CoT	0.3430	0.5159	↑ 0.0623	↑ 0.0528
Qwen3-8B	Naive RAG + IRAC-CoT	0.3223	0.5312	↑ 0.0830	↑ 0.0375
Qwen3-8B	G-Retriever + CoT	0.3375	0.5074	↑ 0.0678	↑ 0.0613
Qwen3-8B	G-Retriever + IRAC-CoT	0.3347	0.5331	↑ 0.0706	↑ 0.0356
RAG + ExpertCoT					
Qwen3-8B	Naive RAG + ExpertCoT	0.3582	0.5280	↑ 0.0471	↑ 0.0407
Qwen3-8B	G-Retriever + ExpertCoT	0.3555	0.5415	↑ 0.0498	↑ 0.0272
ExpertGraphRAG + standard CoT					
LEGO	CoT	0.3555	0.5312	↑ 0.0498	↑ 0.0375
LEGO	IRAC-CoT	0.3734	0.5456	↑ 0.0319	↑ 0.0231
ExpertGraphRAG + ExpertCoT					
LEGO	**Full system**	**0.4053**	**0.5687**	**–**	**–**
Gold article					
Gold Article	+ CoT	0.3472	0.5257	–	–
Gold Article	+ IRAC-CoT	0.3790	0.5468	–	–
Gold Article	+ ExpertCoT	0.4122	0.5894	–	–

Δ = full LEGO minus the row; darker green means a larger gap.



- **ExpertGraphRAG**: +1.25 pp with plain CoT, +4.71 pp with ExpertCoT
- **ExpertCoT**: +1.52 pp with naive RAG, +4.98 pp with ExpertGraphRAG

Results: open-ended legal tasks and retrieval quality

LexRAG_Civil · multi-turn consultation

Metric	Naive	RAPTOR	G-Retr.	Hippo2	LightRAG	LEGO
Factuality	3.707	3.564	3.829	4.329	4.121	**4.486**
Satisfaction	3.829	3.814	3.986	4.300	4.043	**4.600**
Clarity	6.029	6.029	6.136	6.586	6.286	**6.879**
Coherence	4.814	4.743	4.921	5.329	5.257	**5.700**
Completeness	4.193	4.129	4.307	4.686	4.279	**5.093**
Overall	4.314	4.293	4.450	4.829	4.543	**5.164**

PLawBench_Civil · practical case analysis

Metric	Naive	RAPTOR	G-Retr.	Hippo2	LightRAG	LEGO
Reasoning	46.50	44.70	43.00	45.30	43.53	**47.40**
Overall	59.05	58.02	57.22	57.93	58.43	**59.68**

All RAG systems share the Qwen3-8B reader and the Civil Code corpus.

Retrieval quality on LawExamQA_Civil

Method	R@8	R@20	Acc	GCR
Naive RAG	27.99	50.52	28.91	0.61
BM25 (Civil Code)	45.55	57.90	29.05	0.64
Dense (Civil Code)	68.36	79.19	29.60	0.73
LEGO (ExpertGraph)	**72.53**	**81.07**	**30.29**	**0.85**
Gold articles	100	100	31.12	1.00

Recall@k (%) of gold provisions; QA with a zero-shot reader on the top-8. GCR = share of the accuracy gap between zero-shot and gold articles that is closed.

Best on every LexRAG dimension and on both PLawBench scores; the ExpertGraph closes 85% of the gold-article gap.

Conclusion: what we have seen

Motivation	ExpertGraphRAG	ExpertCoT	Results
Textual similarity is not legal relevance; LLMs lack robustness in complex legal reasoning	Expert-built Civil Law graph and NCG: better grounding, structural relations kept, relevant provisions linked	Provision, fact, conclusion: more accurate and consistent multi-hop reasoning	40.53% with an 8B reader, stable up to ≥4 hops, best on open-ended tasks; the two modules are synergistic

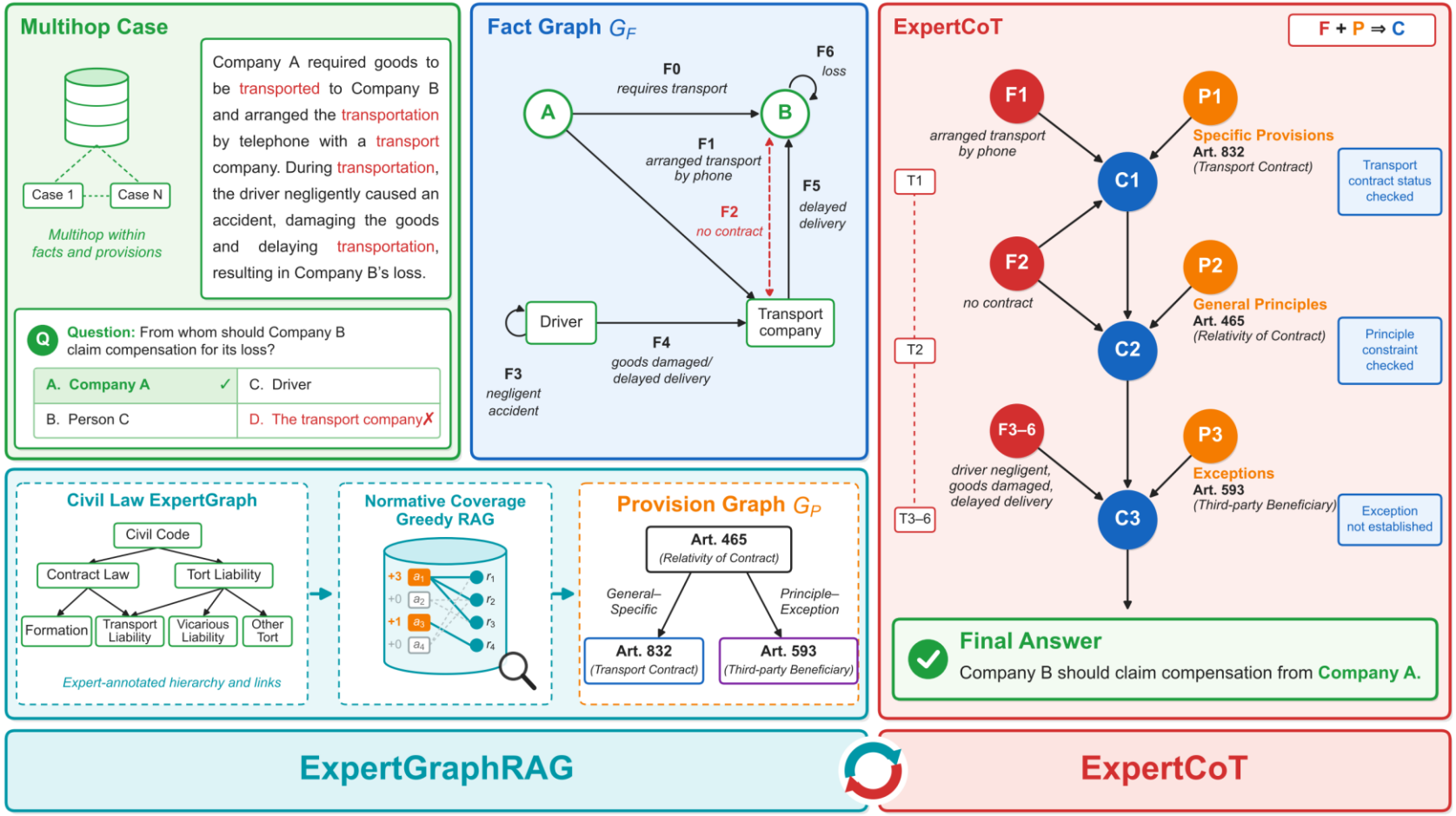
Take-away Injecting expert legal structure into both retrieval and reasoning improves the complex legal reasoning ability of LLMs.

Future work:neuro-symbolic AI challenges



Denotic logic Defeasible logic Non-classic logic	Explicitly modeling the logic and normative relations	Prompt guide	SMT, SAT

Thanks for listening!



Code & data github.com/BLK-WHT/LEGO **Contact** qingjing.chen2@unibo.it

Backup: NCG retrieval algorithm

1	**Route**	top-8 rule units for the query, weights w_r
2	**Support**	g(a) and Γ(a) from the rule–article support edges
3	**Pool**	dense top-50 ∪ graph top-30 articles
4	**Greedy**	add argmax Δ(a | S) until |S| = K = 8
5	**Output**	provision subgraph G_P, passed to ExpertCoT as major premises

$\lambda_{align} = 0.55 \cdot \lambda_{cov} = 0.18 \cdot \lambda_{red} = 0.06$